

IDRIS RAYAN CHAKIB

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EDUCATION

University of M'Hamed Bougara Boumerdes

Boumerdes, Algeria

Master of Science in Software Engineering and Data Processing

Sep 2024 – June 2026

- **Thesis Title:** "AI Pipeline for Genetic Variant Analysis and Virtual Therapeutic Screening" (In partnership with Institut Pasteur d'Algérie).
- **Academic Distinctions:** Thesis Grade: 18/20 (Graduated with Highest Honors / Congratulations of the Jury).
- **Startup Path (Decree 1275):** Developed the project's core framework as an innovative startup. Startup Grade: 19/20 | Awarded the Official National Startup Certificate.
- **Key Coursework:** Network and communication protocols, Cybersecurity, Distributed Systems, Computer Architecture, Data Mining.

University of M'Hamed Bougara Boumerdes

Boumerdes, Algeria

Bachelor of Science in Software Engineering and Information Systems

Sep 2021 – June 2024

- **Academic Standing:** Class Rank: 39 / 182 (Top 21.4% of the cohort / Equivalent to ECTS Grade B).
- **Capstone Project Title:** "Fraud Monitoring Web Solution with Machine Learning" (In partnership with Ooredoo Algeria).
- **Project Distinction:** Grade 16/20.
- **Key Coursework:** Object-Oriented Programming, Calculus, Linear Algebra, Probability & Statistics, Databases.

RESEARCH PUBLICATIONS

Benhenni O., Gaceb D., Touazi F., Rouibi B., Ismail A., Kebour W., **Idris R. C.**, Galleze A., Benidir M. **"SmartADN: An End-to-End Deep Learning Pipeline for Variant Pathogenicity Classification and Mutation-Aware Therapeutic Screening in the North African Context."** Submitted to the IEEE International Symposium on Innovative Informatics of Biskra (IEEE ISNIB'26), 2026. (Paper ID: 101 Currently Under Peer-Review / Forthcoming).

Benhenni O., Gaceb D., Touazi F., Kebour W., **Idris R. C.**, Rouibi B., Ismail A. **"An End-to-End Approach for SNV Pathogenicity Prediction via Progressive Fine-Tuning of Nucleotide Transformer-v2."** Published and presented at the 3rd National Conference on Applications of Artificial Intelligence (A2I'26), 2026. (Official Certificate of Participation Awarded as Presenting Author).

PROFESSIONAL EXPERIENCE

Institut Pasteur d'Algérie

Algiers, Algeria

Data Processing & AI Researcher Intern / Computational Biology

Feb 2026 – June 2026

- Co-designed SmartADN, an integrated pipeline coupling a sequence-only pathogenicity classifier with a mutation-aware therapeutic screening module.
- Fine-tuned the Nucleotide Transformer-v2 genomic foundation model (500M parameters) using a layer-wise unfreezing strategy under a strict gene-split out-of-distribution (OOD) protocol.
- Achieved a near-perfect AUC-ROC of 0.987 and 94% overall accuracy on entirely held-out clinically critical genes (BRCA1, BRCA2, TP53) without relying on population allele-frequency annotations.
- Engineered the protein backbone integration for AEGIS-GT, mapping conformational perturbations via the pointwise difference of ESM-2 embeddings ($\Delta e = e_{MUT} - e_{WT}$).
- Implemented a bi-phasic training strategy using a linear zero-initialised adapter to successfully bridge biophysical Silver-Standard descriptors with experimental Gold-Standard affinity data.

Ooredoo Algeria

Algiers, Algeria

Software Engineering Intern

Feb 2024 – June 2024

- Ingested and preprocessed large-scale Call Detail Records (CDRs) datasets to extract telecom fraud patterns, optimizing data workflows for real-time anomaly detection.
- Designed the end-to-end distributed system architecture using UML diagrams (including deployment diagrams) to ensure scalable infrastructure deployment.
- Developed a secure, full-stack web platform with Role-Based Access Control (RBAC) to streamline analytical workflows for administrators and analysts.
- Built and deployed a feedforward neural network model for fraud detection, validating infrastructure performance with complex real and simulated data streams.

LEADERSHIP EXPERIENCE

SmartADN Startup Project <i>Co-Founder & Lead AI Engineer</i> - Selected for the prestigious Algerian National Student-Entrepreneur program under Ministerial Decree 1275, demonstrating academic and technical leadership. - Architected and developed AEGIS-GT, a custom deep learning model designed for advanced genomic analysis and core startup technology. - Successfully pitched the technical design and AI model architecture to academic committees, securing the official National Startup Certificate.	Boumerdes, Algeria <i>Sept 2025 – Present</i>
NASA International Space Apps Challenge <i>Team Leader & Core Developer</i> - Led a multidisciplinary team of student engineers and peers during the 48-hour global hackathon to address complex astronomical challenges. - Co-designed and engineered a 3D interactive web-based Orrery application to simulate celestial body trajectories using open-source NASA telemetry data. - Managed project workflows, delegated tasks among university peers under strict time constraints, and delivered the functional backend framework, awarded the Galactic Problem Solver Certificate.	Global / Algiers Cohort

PROJECTS

Multi-Robot Autonomous 2D Swarm Exploration - Engineered a multi-robot swarm mapping system from scratch using a continuous Teacher-Student reinforcement learning strategy paired with the A* pathfinding algorithm. - Coded the entire Deep Q-Network (DQN) architecture, backpropagation loops, and neural layers using raw Python/NumPy.	
Nexus: Decentralized P2P Encrypted Chat System - Designed a robust peer-to-peer distributed framework featuring dynamic private chat rooms, logical isolation, and multi-threaded file transfer protocols. - Implemented physical concurrency controls and data consensus layers using Lamport Logical Clocks, custom Byzantine-resilient voting systems, and multi-layered Fernet cryptographic encryption via Tkinter GUI. - Dealt with complex infrastructure stacks involving distributed systems, concurrency control, and socket-based network protocols.	
Moogole: Natural Language Processing Search Engine - Built a highly optimized localized text search engine integrating foundational Natural Language Processing (NLP) preprocessing pipelines. - Developed custom algorithmic structures for tokenization, automated stop-word elimination, Porter-style stemming/unstemming, and contextual Bi-gram language modeling.	

ADDITIONAL CREDENTIALS & ACTIVITIES

Professional Certifications Machine Learning Specialization (DeepLearning.AI): Certified core expertise in neural networks and AI system design. Google Data Analytics Professional Certificate: Certified proficiency in data cleaning, SQL analysis, and problem-solving.	2025 – 2026
Active Competitor, Codeforces Platform Current Rating: 1166 (Actively progressing toward top-tier algorithmic milestones via timed challenges.) Ranked 273rd out of 3,830 active global participants (Top 7%) in the ICPC Online Challenge 1 (powered by Huawei); engineered a custom scheduling heuristic to optimize concurrent request routing for Edge-Cloud Collaborative LLM Inference loops.	2026 – Present

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS	
Cloud Platforms: Google Cloud Platform, AWS, Microsoft Azure	
Frameworks: Next, Express, FastAPI, Flask, Pytorch, Keras, Scikit-learn, Tensorflow, Hugging Face (Transformers)	
Developer Tools: Linux/UNIX, Git, Docker, Kubernetes, Node.js, VS Code, Visual Studio, Pycharm, IntelliJ, Eclipse	
Libraries: REST APIs, Microservices, React, pandas, Numpy, Matplotlib, seaborn	